

Theory of Computation

Relation Machine will do computations on the basis of Theory

Definition: Theory of Computation is branch of computer science that deals with how efficiently problem can be solved on a model of computation using an algorithm.

Theoretical computations is built upon two foundational elements: (i) Principles (ii) Concepts

(i) Principles: The underlying rules, mathematical logic, formal laws and proofs. (e.g. set theory, formal languages)

(ii) Concepts: Abstract ideas and models (e.g. state transitions, decidability) etc.

Every Computational machine is defined by four Core characteristics: (i) Compatibility (ii) Limitations (iii) Complexity (iv) Performance.

(i) Compatibility: How different machine models equate to or convert into one another.

(ii) Limitations: What a specific machine model can or cannot compute/recognize.

(iii) Complexity: The amount of resources (time and space/memory) a machine consumes for an input size n .

(iv) Performance: How efficiently or conveniently a machine executes computation.

Application of TOC

(i) It plays an important role in compiler design.

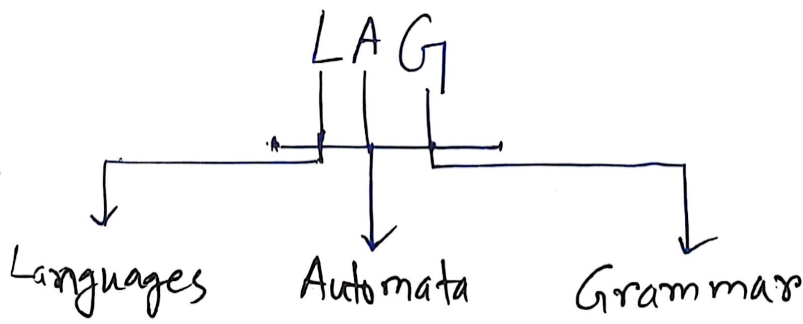
(ii) In switching theory and design and analysis of digital circuits automata theory is applied.

(iii) Design and Analysis of Complex hardware and software system.

iv) To improve correctness of the program automata theory is used.

v) To design finite state machines such as ~~more~~
moore and mealy machines

Pillars of TOC:



Language

to make language we need →

- Symbols
- Alphabets
- Strings

Symbols: Digits, letters
0, 1 a-z

Def: Basic building block of a language or smallest unit of a language are called Symbols

Alphabets: Finite set of symbols are called Alphabets. It is denoted by $(\Sigma) \rightarrow$ sigma

For example: $\Sigma = \{a, b\}$, $\Sigma = \{0, 1\}$

Strings: Finite/Infinite sequence of symbols are called Strings. It is denoted by $(w) \rightarrow$ omega

Examples: $\Sigma = \{a, b\}$ and string length = 2, strings?

$\therefore w = \{aa, bb, ab, ba\}$ [finite sequence]

formula $a \rightarrow \cancel{(\Sigma)} \left[\begin{array}{c} \text{string length} \\ \Sigma \end{array} \right]$

length of string denoted as $|w|$

Ques: $w = \{abaabab\}$ sequence = 1.

$|w| = ?$ length = 6

Language: Collection of all possible strings are called Language. For example: ~~[If]~~ All C-programms are strings. And If it is valid then that is Language

Power of Sigma (Σ) (alphabets)

Lecture 3

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$\Sigma^0(\Sigma)$ epsilon

$$\Sigma^1 = \{a, b\} / \{0, 1\}$$

$$\Sigma^2 = \{aa, bb, ab, ba\}$$

$$\Sigma^3 = \{ \dots \}$$

$$\Sigma^K = \{ \dots \}$$

mother of all strings

formula

$$\Sigma^{\text{str-length}}$$

⇒ If sigma is an alphabet, we can express set of all strings of certain length from that alphabet by using exponential notation,

It is denoted by Σ^K

K is the set of strings length

$$\Sigma = \{a, b\}$$

$$L = \{w \mid w, \text{ where each string starts with 'a'}\}$$

$$= \{a, aa, aaa, ab, \dots\}$$

↓ Infinite lang.

Finite Representation

Finite Automata

FR

Yes → [a, aa, ...]

No [baa]

Automata theory and Language

It deals with definition and properties of various mathematical models of computation.

Definition: Automata is a self operating system on machine that performs predefined sequences of operation automatically.

Mathematical Models: (i) finite Automata (ii) Context Free Grammar

(iii) Turing Machine.

Types of Automata:

- (i) Finite Automata (FA)
- (ii) Push down Automata (PDA)
- (iii) Linear Bound " (LBA)

Finite Automata: F.A. is an abstract computing device. It is a mathematical model of a system with discrete inputs, outputs, states and set of transitions from state to state that occurs on input symbols from alphabets.

representation: (i) Graphical { transition diagram or transition table }

(ii) tabular { transition table }

(iii) Mathematical { transition function or mapping function }